

Revealing a Concealed Eye Disorder in Children: Designing a Solution Grounded in HCI Principles

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Abstract

Intermittent Exotropia (IXT), an eye condition where one eye deviates outwardly, is complex to diagnose in young children. Observation of its manifestation relies on several factors, with the key one being inattention. My goal is to develop a method to diagnose IXT in kids at home or a clinic using a phone or tablet. My work starts with systematically reviewing publications claiming to use gamification for medical practitioners. This then motivates the use of a game design technique called “Juicy Design”, which I examine as a method to induce low attention and disengagement. Next, I will work on a method to accurately measure eye deviations using computer vision, a dataset made of original and synthetic images, and a model to generate relevant IXT measurements, which will be used in the app that displays cartoons processed with different levels of Juicy design to induce IXT.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Laboratory experiments**; **Empirical studies in HCI**; *Interaction techniques*.

Keywords

inattention, children, experiments, gaze tracking, psychophysiology

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1 Introduction

Intermittent Exotropia (IXT) is a sub-type of strabismus (eye misalignment), which manifests with an eye moving “outwards” at irregular intervals [39]. Children with strabismus are at increased risk for irreversible vision loss, while adults carry an increased risk of injury [5]. Children who grow up with symptomatic undiagnosed, and untreated oculomotor disorders are more prone to develop behavioural and emotional issues [6], impacting them and

their caregivers with long-term worry [23]. The lifelong impact of Strabismus is a public health concern [47], and this is our primary motivator to create a solution to aid patients, their caregivers, and eye doctors. Diagnosing eye-related oculomotor disorders is complex, especially with children. An optometrist or ophthalmologist with access to the necessary equipment, knowledge, and skills is the minimum anticipated condition for a diagnostic exam. Prohibitive cost or wait times can make these resources much harder to access, especially in rural areas, remote indigenous communities, developing nations. Even countries with socialised medicine, eye care access can be poorly integrated or nonexistent [37]. With IXT, diagnosis is even more challenging due to when it does, or does not present. External factors that reveal IXT include sunlight [32, 51] and panoramic viewing [51]. Experiential circumstances also reveal IXT, including inattention [32, 39], stress, and fatigue or illness [32]. The latter factors relate to when the patient is experiencing *inattention or disengagement*.

Our approach is to combine “just good enough” eye tracking available on current smartphones with a child-oriented application that leverages suitable design techniques, making diagnosis more reliable and predictable. My work starts with systematically reviewing publications examining the general use of gamified applications for medical practitioners. Motivating the use of gameful design technique called “Juicy Design”, which I examine as a method to induce low attention and disengagement. We then, develop user and engineering requirements to design a mobile application suitable for medical practitioners and caregivers. Next, we develop a method to accurately measure eye deviations using computer vision, dataset generation and a model trained to achieve IXT deviation measures.

By the end of my PhD, I intend to achieve four objectives: (1) validate that visual embellishments can influence a person’s attention and engagement, as well as measure its impact; (2) create a technique to alter video content that elicits a state of low attention or disengagement in young patients; (3) develop a set of engineering requirements and user requirements for a solution suitable for use by optometrists, caregivers, and children; and (4) create a dataset and deep-learning model that transforms the camera of a mobile phone into a dependable IXT measurements tracker.

2 Related Work

2.1 Intermittent Exotropia

Strabismus is an eye condition where a person’s eyes are not working in tandem, and visibly not parallelly aligned. Different classifications are given based on the direction in which the uncooperative

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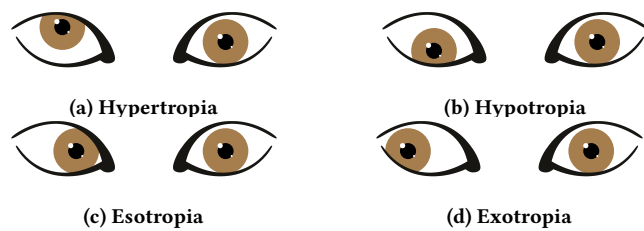


Figure 1: Different strabismus eye deviation orientations, we will focus on (d) Exotropia in this paper.

eye deviates. Figure 1 shows each type of eye deviation. It is estimated that close to 150 million people in the world are living with some form of strabismus [5].

IXT is a subtype of strabismus that falls under the exodeviation label (Figure 1d), it represents 75 to 90% of Exotropia cases [18, 38]. IXT first manifests in children under five years old [13], and it is characterised by irregular outward deviation of one eye, typically with both eyes affected. Pan et al. [38] explored the epidemiology of IXT in preschool children aged 3 to 6 years in China, finding that 1 in 30 children are affected by it. A study focused on the white population in the United States estimated there will be approximately 43 thousand new cases of exotropia in children each year [18]. IXT is also referred to as distance exotropia, divergence excess exotropia, periodic exotropia, and exotropia of inattention [12].

In most cases, IXT becomes visible in babies aged 6 to 12 months [31]. Parents may visually identify the deviation in their children during early childhood or late infancy and report that it happens more often when their child is tired, inattentive, sick, or daydreaming [15, 53]. The degree of deviation typically classified as simply "small" or "large". Small degree IXT has a higher potential to cause visual symptoms, especially if presenting intermittently [43]. It is likely to be missed or neglected to be addressed, enabling disease progression, and increasing the likelihood of impairment [1].

Children with IXT are impacted in several ways, including visual, motor, and academic issues. They may suffer from eye strain symptoms, blurred vision, headaches, visual disarray, and difficulties when reading [28, 42]. Motor issues are manifested through locomotion difficulty like persistent stumbling and walking into walls. It also reduces eye-to-hand coordination, affecting gross motor movements like sports performance [44] and fine motor skills like artistic expression and handwriting [41].

Historically the diagnostic criteria and comparison of treatments of IXT have been controversial and uncertain [14]. Previous research reported conflicting results, which can be attributed to a lack of methodology rigour producing biased results [22]. Economides et al. [14] argues that the body of research published may be insufficient to provide individual future expectations of the progression of IXT because each patient's oculomotor evolution depends on their personal circumstances, treatment choices, lifestyle, amongst many other factors. Diagnosis of IXT can be complex because it does not have a reliable manner to obtain reliable measures of its necessary parameters, and depends on several factors. For example, the patient's cognitive attention level [15, 28, 39], the target used for fixation examination, the distance between the patient and the target, and the patient's developed suppression mechanisms.

The two parameters necessary to diagnose and determine the severity of IXT are (1) the angle of deviation, referred to simply as *magnitude*, and (2) the proportion of time the patient's eye spends deviated in a 24 hour period, referred to as *rate of occurrence* [4]. While the measurement of magnitude is straightforward to measure if manifested in the optometrist's office, a *cover test* is the alternate way to obtain a measurement of non-manifested deviation. The *cover test* uses an occluder paddle, measuring the time taken to regain alignment for each eye, this method is still not proven to correlate with spontaneous exotropia manifestation [14], nevertheless it is widely used. A patient can avoid displaying their eye deviation even when an optometrist performs consecutive cover tests [28], compelling the doctor to rely on a parent's observation, recall, quantification, and recount of their child's eye deviation, which has been reported to be unreliable [14]. Lastly, *Index Tests* are used to detect strabismus in screenings performed by non experts in communities. These are divided in different types, in case of strabismus and IXT we would need the tests which directly identify ocular misalignment: first we have corneal reflections tests such as the *Hirschberg* and the *Krimsky*, and fundus reflections test like the *Brückner* [26]. The rate of occurrence (2) to this date has no solution for obtaining a precise proportion of the deviations a patient experiences [24]. Although different methodologies are used to assess the control a patient has over their IXT, these are qualitative in nature and their result is given based on a numerical scale, which does not determine the quantity of time spent in deviation, in addition scores vary when assessment is performed several times in a day [2].

2.2 Optometry Diagnostic Novelty Overview

In 2008 Carvelho et al. [8] predicted that eye therapy would be enhanced by games and technology. A bibliometric investigation published in 2023 analysed the past two decades of research on Strabismus and Artificial Intelligence (AI) has shown that 59.12% of the articles were published in the past five years [56].

The research published is either focused on diagnostics or treatment of eye conditions. Diagnostics are the focus of this project, therefore we will discuss past work that contributes to its progress.

Initially, eye health screening technologies based on photos yielded a high rate of false positives [40]. The next innovations researched solutions for diagnostics that required top-of-the-line expensive devices equipped with eye tracking like Tobii [3] and VR headsets [34], which produced reliable diagnostic results [10, 54] and a dataset representative of persons of normal and strabismic vision [9]. Research that has utilised off-shelf mobile devices reported successful implementation of picture to diagnosis systems [11, 17], promising deep learning models that use eye-specific features aimed at improving detection/prediction of eye disorders [46, 55]. Zou et al. [58] examined the reliability of such systems by checking their ability to get reproduced, and the results showed higher repeatability for the eye tracking based technologies compared to the traditional methodologies. The body of research in this area only focuses on the technological facet of a diagnosis, neglecting the known weaknesses of the diagnostic process.

Two main themes are present in this body of work: (1) the need for obtaining reproducible measurements and be more exact in the process of strabismus diagnostics; (2) the wish to make these

more exact systems available to laypersons, and ensure children have access to a diagnosis. Despite acknowledging the need for a more child friendly approach [55], and a methodology that enables the child to experience inattention or disengagement [15, 28, 39], to my knowledge, none of the solutions attempts to tackle this issue. My work focuses on designing an experience that intends to explore both facets of this challenge, the child's experience and the detection of IXT with a mobile phone or tablet camera.

3 Proposed Solution and Methodology

This project seeks to develop a cutting-edge solution, spanning multiple crucial HCI research areas covering Child-Computer Interaction fundamentals, affective computing theories and engagement strategies, visual embellishment principles, and vision optometry fundamentals. Our system will be low-cost, intuitive, and non-intrusive to children.

3.1 Project 1: Systematic Review: Games for Medical Practitioners

Here we investigate the state of the literature published about systems that use gamification or are games designed for medical practitioners (MPs) as main users. A review published, with a similar goal to ours and that targeted medical focused databases resulted in a lack of reporting of data that is necessary for the advancement of HCI and Health; such as findings that report on MPs perspectives and considerations towards their job, in addition to the lack of information the quality of the data was lacked according to the authors [45]. Previous HCI-Health systematic reviews acknowledge the importance of focusing on the correct databases in order to find HCI-relevant data [21]. Our research question (RQ) is:

RQ1: *What kinds of game or gamification elements are being used for the context of supporting MPs?*

We followed the convergent-integrated JBI methodology for mixed-methods systematic reviews (MMSR) using thematic summaries for qualitzation [52]. The MMSR is used when research questions can be answered through quantitative and qualitative methods alike, we extracted textual passages and quantitative data that matched fields present in our best fit framework [7] that was composed of the JBI data extraction tool form [35], the Hexad [49] and Elements of Gameful Design frameworks [48], as described in detail below. The best-fit methodology is often used to inform policy and practice of healthcare information synthesis [35]. We developed and registered our review protocol on the Open Science Framework (OSF) [16], which we don't share here due to the manuscript being under peer-review. The databases used for our search methodology focused on HCI practice primary publishing venues.

Our analysis shows that: (1) There is a shortage of games developed exclusively for MPs, specifically beyond the patient-doctor relationship. Thus, there may be a research opportunity to leverage games and gamification tailored towards MPs to support them in their work. (2) Existing innovations and artifacts designed for MPs are sparsely reported on, and those that did include minor details, do not draw on gameful design principles and processes. This suggests there is an untapped potential of creating more meaningful gameful innovations and generating literature that contributes to understanding specifically what game design approaches work best

for MPs and their varied specialisations. Our work contributes to the area of HCI research in games by identifying these underexplored research opportunities. Based on this, we draw implications for what appear to be the key relevant challenges to designing and deploying games and gamification systems specifically centred around MPs. This lays the foundation for future researchers to explore which design factors can support acceptance of games and gamification systems among MPs.

3.2 Project 2: Intermittent Exotropia Requirements Engineering

According to Karlsson and Ryan [30], Requirements Engineering (RE) must aim for clear and unambiguous requirement priorities, which will enable a focus on the project and the system that one aims to build. The purpose of this project is to identify a strategy to reach the following goals:

1. Trigger the eye deviation in toddlers and children;
2. Measure the eye deviation angle;
3. Keep track of a child's eye deviation control;

Zimmermann and Grötzbach [57] created an approach to integrate user centred design (UCD) to RE practices and process, which aids in keeping the planning, tracing, and execution in sync, making the process more manageable. The backbone of a good RE process practice is to be human focused. Hidellaarachchi et al. [25]'s systematic review identified human aspects that are typically considered and were impactful for successful RE. Four human aspects were identified; Communication, personality, motivation, and gender. In addition, it was discovered that investigating two or more human aspects reveals relationships and how they impact the RE process. The similarities of RE and HCI practices have been adopted and merged to achieve success when building systems that require both facets [27]. According to Niknafs and Berry [36] a team that is made of domain experts and domain ignorants is more effective when developing solutions. Therefore, this project will take advantage of the newbie effect by consulting with medical practitioners, the literature, patients and caregivers to ensure all human aspects uncovered are taken into account.

We consulted with a practising optometrist who has provided us with relevant literature on IXT. In addition, we performed a simple literature review and reported symptoms and cues, the impact, presentation, diagnosis, and guidelines to determine severity. Next, we define our Requirements Engineering (RE). Our objective is to obtain clear and unambiguous requirement priorities [30], which will enable us to achieve the objective of triggering the eye deviation in toddlers and children. Using the model from Gunter et al. [20], we look at the hardware and software as parts that can help us build a system, which can be constructed by a combination of any social-technical artefact.

3.3 Project 3: Disengagement in Adults: Juicy Design & Eye Tracking

Recall that Juicy Design is typically used in games and involves adding non-functional visual embellishments to increase engagement. This project investigates if too much or too little juicy design

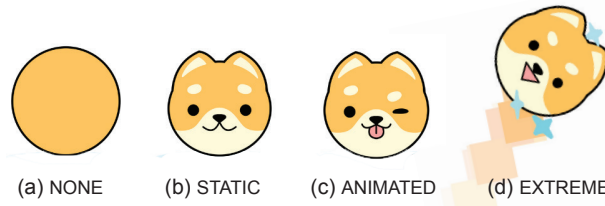


Figure 2: Target stimulus levels: (a) NONE; (b) STATIC; (c) ANIMATED; (d) EXTREME.

can lower attention of adult participants during a controlled eye tracking task. Our RQ is:

RQ1: *Can Juicy Design affect attention levels in adults during an eye tracking experiment?*

The goal of this experiment was to examine the effect of the level of juicy design on attention. A total of 16 participants completed standard smooth pursuit and saccade eye-tracking tasks, with interleaved (non-task) rest breaks. Pupil dilation and blinking were logged as objective metrics that gives us cues on one's level of attention. After both parts of the experiment were complete, the participant completed a short questionnaire about their general impression of the experiment and specific questions about the four levels of juiciness. Lastly, we conducted a short semi-structured interview. This was a within-subjects design with one independent variable representing JUICINESS with 4 audio-visual levels: NONE, STATIC, ANIMATED, EXTREME (Figure 2). The order for JUICINESS was counter-balanced using a Latin square. We used a fixed order of

tasks defined by ascending level of difficulty and mental effort, because previous literature claims that this standard results in reliable pupil diameter measurements for both within- and between-subject experiments [29].

Our results contribute evidence that *too much or too little juicy design can negatively affect attention* and, subsequently, reduce engagement with an audiovisual stimulus. Our most effective use of juicy design levels was when EXTREME and ANIMATED were employed on a task, and followed by boring juicy break (Figure 3, Figure 4). Our empirical results further demonstrate the general efficacy of visual embellishments, and they suggest new ways it can be used for purposes other than enhancing gaming experience and attention. An application I aspire to see or explore in the future with these findings is to induce disengagement in order to nudge a user to put their device down and take a break. This could curb screen addiction or help enforce screen time limits for children.

This project was published in the Proceedings of the 2024 International Conference on Advanced Visual Interfaces [33].

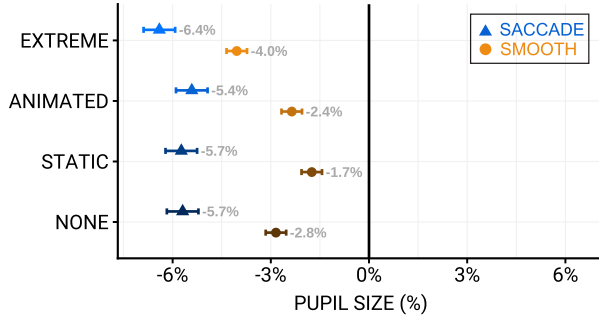


Figure 3: Gaze task *Pupil Size* by JUICINESS for Smooth Pursuit and Saccade tasks (error bars in all graphs are 95% CI).

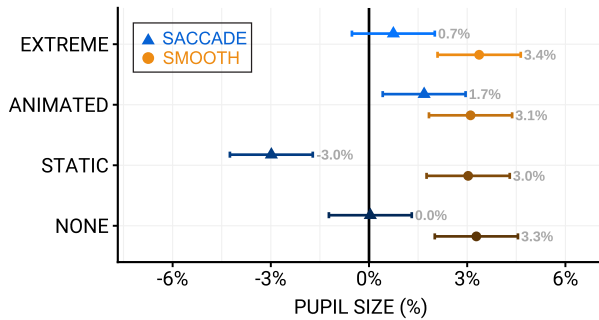


Figure 4: Rest Breaks *Pupil Size* by JUICINESS after Smooth Pursuit and Saccade tasks.

3.4 Project 4: Disengagement in Children: Juicy Cartoons

This project investigates the impact of different levels of Juicy Design on children's attention and engagement while watching a cartoon episode. I am particularly interested in exploring how juiciness can induce states of low attention and disengagement. The protocol consists of an adapted version of the "look" task, which was part of an attention study investigation with infants [19]. I measure their attention and engagement using qualitative video coding the child's behaviour, and by gathering data on pupil size, proportion of observation in area of interest, and blink rate. Our RQ is:

RQ1: *Does the level of Juicy Design use in a cartoon episode impact attention and engagement of children?*

This work is currently in progress, lab studies have been performed with 11 of 35 children.

3.5 Project 5: Oculomotor Measurements Data Set and Model

This project lays the technical foundations for phones and tablets to act as an oculomotor tracker for IXT relevant movements. This project is divided into two parts: the first focuses on creating a dataset of diverse images of people exhibiting highly controlled IXT eye movement deviations. The second one focuses on using the dataset with deep learning techniques to create a bespoke model for the identification of IXT. I investigate the following question:

RQ1: *Can we enable a mobile phone camera to perform "just good enough" IXT detection with a bespoke dataset and deep learning model?*

To create the mobile application, we need a method to accurately measure eye deviations using computer vision. We will use an automated data-gathering protocol to collect a large dataset of images with controlled head angles and eye gaze deviations under different lighting conditions. We use this dataset of healthy users to generate a synthetic dataset of images exhibiting controlled IXT characteristics and use this to train a model to generate relevant IXT measurements. My goal is to use this model in an app that displays cartoons processed with different levels of Juicy design to induce IXT.

The goal of this project is to enable phones and tablets to track oculomotor movements. Previous work that explored the use of off-the-shelf mobile devices used traditional methods for diagnosis such as the Hirschberg test [11, 17], and limited their algorithm calculations to one measure only. Works that focused on creating deep learning models aimed at improving the detection or prediction of strabismus mostly used a convolutional neural network (CNN) or adaptations of the Hirschberg test, which does not produce a reliable diagnosis. Zhang et al. [55] developed a multi-feature fusion model (MFFM) that outperformed single feature models in strabismus detection. Another issue is these prior works do not publish model training code [50, 55].

3.6 Project 6: Future Work

Projects 2 (3.2), 3 (3.3) and 4 (3.4) define the pillars to build an application suitable for all potential users, grounded in HCI principles and techniques. In parallel, Project 5 (3.5) enables mobile phones to act as a gaze tracker.

This project (3.6) brings together all the knowledge gathered throughout my PhD and tests a functional prototype capable of performing a diagnosis in a lab setting using the results of all previous projects. I combine the Juicy Cartoon display and IXT measurement collection tool, and run a trial test to informally validate our success.

4 Author's PhD Status

I am a third-year Ph.D. student in Computer Science, at the University of Waterloo, ON, Canada. I am advised by Dr. Daniel Vogel and have the support and Co-Supervision of Dr. Ben Thompson (School of Optometry and Vision Science, University of Waterloo). I spent the first two years of my PhD getting familiar with the relevant Optometry knowledge, and Computer Science needs due to my unconventional background. I have passed both my PhD Comprehensive Examinations, being in the phase of PhD Candidacy. The Comprehensive Examination I requires PhD students to complete high level courses in a breadth of topics in computer science, I achieved a GPA of 92%. The Comprehensive Examination II is held by a committee and chairs who evaluate a written thesis proposal, followed by a presentation, questions and discussion. I have never previously attended a Doctoral Consortium, or any CHI conferences in person. I strive to work in Academia in the future,

and hope that I can succeed in switching from industry to this fulfilling career that contributes to education and scientific evolution in society.

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